Siva Sreedhar

120

$2 \times (15 + 45) / 100$

COURSE: CHE 312: Transport Phenomena II

Spring Semester 2020

Department of Chemical Engineering

University of Illinois at Chicago

Midsemester Exam #1

Only one sheet of paper 5 x 7 is allowed

Duration: 50 minutes **--- February 10, 2020**

Both Problems must be done

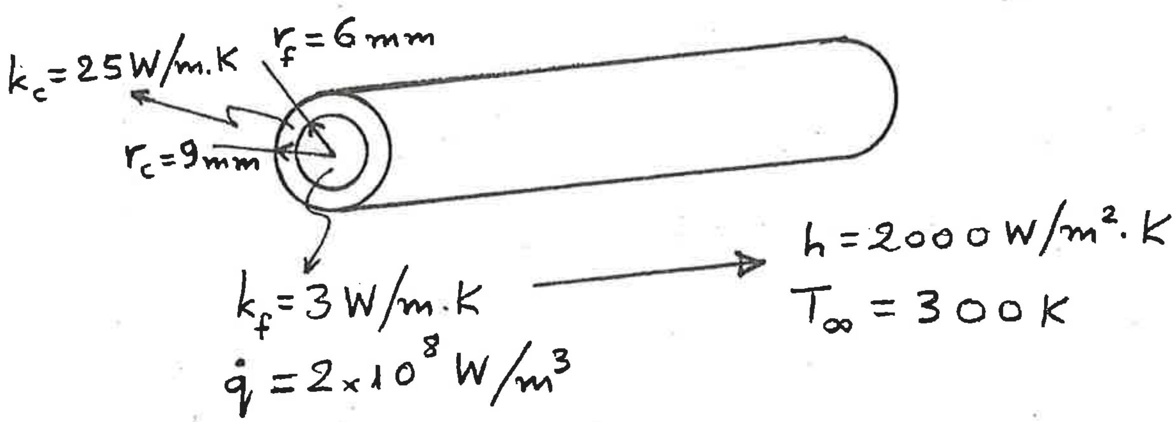
Problem 1: (50 points)

A nuclear reactor fuel element consists of a solid cylindrical pin of radius r_f and thermal conductivity k_f. The fuel pin is in good contact with a cladding material of outer radius r_c and thermal conductivity k_c. Consider steady-state conditions for which uniform heat generation occurs within the fuel at a volumetric rate $\dot{q}$ and the outer surface of the cladding is exposed to a coolant that is characterized by a temperature T_∞ and a convection coefficient h.

$\left(\frac{20+15}{50}\right)$

a) Obtain equations for the temperature distributions $T_f(r)$ and $T_c(r)$ in the fuel and cladding, respectively. Express your results exclusively in terms of the foregoing variables.

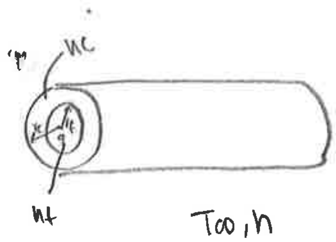
b) Consider a uranium oxide fuel pin for which $k_f = 3$ W/m·K and $r_f = 6$ mm and cladding for which $k_c = 25$ W/m·K and $r_c = 9$ mm. If $\dot{q} = 2 \times 10^8$ W/m^3, h = 2000 W/m^2·K, and $T_\infty = 300$ K, what is the maximum temperature in the fuel element?



1)

Assumptions? Siva Sreedhar

(35)



$\frac{\partial T}{\partial t} = 0$

From $r=0$ to r_f:

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) + \dot{q}_f = 0$$

$$\int \frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) = \int -\dot{q}_f r$$

$$kr\frac{\partial T}{\partial r} = -\frac{1}{2}\dot{q}_f r^2 + C_1$$

$$\int \frac{\partial T}{\partial r} = \int \frac{-\frac{1}{2}\dot{q}_f r^2}{kr} + \frac{C_1}{kr}$$

$$T_f(r) = -\frac{1}{4}\frac{\dot{q}_f}{k}r^2 + \frac{C_1}{k}\ln|r| + C_2$$

Boundary conditions:

Ⓐ $-k\frac{dT}{dr}\Big|_{r=0} = 0$ (symmetric)

Ⓑ $T(r_f) = T_1 \} = T_c$

Missing B.C.'s
a) at $r = r_f$
b) at $r = r_c$

Not a B.C. → Also: $\dot{q}(r_f/2) = k\frac{(T_1 - T_2)}{(r_c - r_f)} = h(T_2 - \infty)$

$\dot{q} = W/m^3$

volume

$$\frac{\cancel{\pi} r_c^2 \cdot \cancel{L}}{(2\cancel{\pi} r_c)(\cancel{L})}$$

$L_c = r_c/2$

$$\frac{\dot{q}(r_f/2)(r_c - r_f)}{k} = T_1 - T_2$$

$$T_1 = \frac{\dot{q}(r_f/2)(r_c - r_f)}{k} + T_2$$

$$\dot{q}(r_f/2) = h(T_2 - T_\infty)$$

$$\dot{q}(r_f/2) = h\left(T_1 - \frac{\dot{q}(r_f/2)(r_c - r_f)}{k} - T_\infty\right)$$

$$T_1 = \frac{\dot{q}(r_f/2)}{h} + \frac{\dot{q}(r_f/2)(r_c - r_f)}{k} + T_\infty$$

Thus, $T_2 =$

$$T_1 - (\quad) = \frac{\dot{q}(r_f/2)}{h} + T_\infty$$

From $r_f \to r_c$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(kr\frac{\partial T}{\partial r}\right) = 0$$

$$kr\frac{\partial T}{\partial r} = C_1$$

$$\int \frac{\partial T}{\partial r} = \int \frac{C_1}{kr}$$

$$T_c(r) = \frac{C_1}{k}\ln|r| + C_2$$

B.C:

$$-k\frac{dT}{dr}\Big|_{r=r_c} = h(T - T_\infty)$$

$\{ T(r_f) = T_1$
$\{ T(r_f, r_c) = T_2$

$$T_1 = \frac{C_1}{k}\ln|r_f| + C_2$$

$$-\quad T_2 = \frac{C_1}{k}\ln|r_c| + C_2$$

$$T_1 - T_2 = \frac{C_1}{k}\ln\left(\frac{r_f}{r_c}\right)$$

$$\frac{h(T_1 - T_2)}{\ln(r_f/r_c)} = C_1$$

Thus, from notecard derivation of in class radial system

$$T(r) = T_{s,2} + \Delta T\frac{(\ln(r/r_2))}{\ln(r_1/r_2)}$$

a) $\boxed{T_c(r) = \frac{\dot{q}(r_c/2)}{h} + T_\infty + \left(\frac{\dot{q}(r_f/2)(r_c - r_c)}{k}\right)\frac{\ln(r/r_c)}{\ln(r_f/r_c)}}$

in Terms of k ?

Plugging in B.C:

Ⓑ $T_1 = -\frac{1}{4}\frac{\dot{q}_f}{k}(r_f)^2 + \frac{C_1}{K}\ln|r_f| + C_2$

$$\Delta T = T_1 - T_2 = \frac{\dot{q}(r_f/2)(r_c - r_f)}{k}$$

Ⓐ $-k\frac{dT}{dr}\Big|_{r=0} = -\frac{\frac{1}{2}\dot{q}_f(0)}{k} + C_1/(0)$

C_1 must equal 0.

$$C_2 = T_1 + \frac{1}{4}\frac{\dot{q}_f}{k}(r_f)^2$$

Thus

a) $\boxed{T_f(r) = -\frac{1}{4}\frac{\dot{q}_f}{k_f}(r_f)^2 + T_1 + \frac{1}{4}\frac{\dot{q}_f}{k_f}(r_f)^2}$

$$\left(\frac{\dot{q}(r_f/2)}{h} + \frac{\dot{q}(r_f/2)(r_c - r_f)}{k} + T_\infty\right)$$

b) now plugging in values:

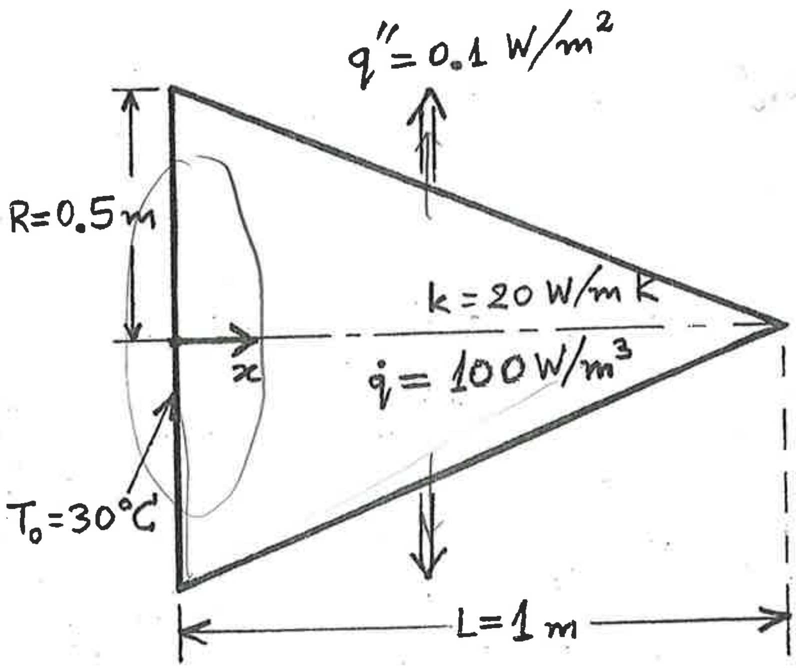
$T_f(r=0) = T_1$

$$T_1 = \frac{(2\cdot10^8\ W/m^3)(6\cdot10^{-3}/2)}{2000\ W/m\cdot k} + \frac{(2\cdot10^8)(3\cdot10^{-3})(3\cdot10^{-3})}{25} + 300K$$

b) $\boxed{T_1 = 672K}$ max temp ✗ Incorrect

Problem 2: (50 points)

A solid cone as shown is of circular cross section and the radius of its base is 0.5 m. The cone of length 1 m and thermal conductivity 20 W/m·K has uniform volumetric heat generation of 100 W/m^3. A constant heat flux of 0.1 W/m^2 is leaving the cone through the lateral surface of the cone, while the end surface at x = 0 is maintained at T_o = 30 °C. Determine the temperature profile T(x) and the temperature T(L).



$\frac{\partial^2 T}{\partial z^2} + \dot{q}/k = 0$

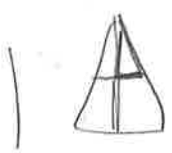
$\frac{\partial^2 T}{\partial z^2} + \frac{\partial^2 T}{\partial r^2} = 0$

$T = T(x)\,T(r)$

$T(x) =$

100 $\frac{\partial^2 T}{\partial z^2} + \dot{q}/k = 0$

25



$T(x) = \frac{\partial^2 T}{\partial x^2} + \dot{q}/k - (.1\ W/m^2)/(Lc) = 0$

$\frac{S}{r} = \frac{x}{1}$

$S = xr$

B.C:

$T(0) = 30°C$

total energy bal.

Energy Balance:

$(100\ W/m^3)(1m) = +k\frac{dT}{dx} + .1\ W/m^2$

$(100\ W/m^2) - (.1\ W/m^2) = -k\frac{dT}{dx}$

$+\frac{99.9\ W/m^2}{k} = \frac{dT}{dx}$

$\frac{S}{r} = \frac{1}{x}$ $V = \frac{\pi r^2 h}{3}$

$Sx = r$

cross sectional area =

$x = 2r$

$\frac{2\pi r h}{3}$

$\frac{(1-x)}{1} = \frac{r}{S}$ $Lc = \frac{\pi r^2 h}{3} \div \frac{2\pi r h}{3}$

$Lc = \frac{\pi r^2 h}{2\pi r h}$ $r/2$

$S(1-x) = r$

$S - Sx = r$

$1 - r/S = x$

$1 - 2r = x$

$(100\ W/m^3)(1m$

now to fix

answer: show

x dependence on changing

r and put area in terms of

this changing value.

$T(L) =$

$\boxed{T(1m) = 34.995°C}$

$\int +\frac{99.9\ W/m^2}{20\ W/mK} = \int \frac{dT}{dx}$

$T(x) = \frac{+99.9\ W/m^2}{20\ W/mK}x + C_1$

@ $x = 0, T = 30$

$T(0) = 30 = C_1$

$\boxed{T(x) = \frac{+99.9\ W/m^2}{20\ W/mK}x + 30}$